

IAS PRELIMS/MAINS



SCIENCE & TECHNOLOGY



A COMPREHENSIVE COVERAGE

MODULE - 8

OUTER SPACE TREATY

OUTER SPACE TREATY

- The Outer Space Treaty, formally the Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, including the Moon and Other Celestial Bodies, is a treaty that forms the basis of international space law.
- Presently, 110 countries are parties to the treaty, while another 23 have signed the treaty but have not completed ratification.
- The treaty is overseen by the United Nations Office for Outer Space Affairs (UNOOSA).



OUTER SPACE TREATY

The Outer Space Treaty represents the basic legal framework of international space law. Some of its principles are:

- It bars states party to the treaty from placing weapons of mass destruction in Earth orbit, installing them on the Moon or any other celestial body, or otherwise stationing them in outer space.
- It exclusively limits the use of the Moon and other celestial bodies to peaceful purposes and expressly prohibits their use for testing weapons of any kind, conducting military maneuvers, or establishing military bases, installations, and fortifications. However, the Treaty does not prohibit the placement of conventional weapons.



OUTER SPACE TREATY

The Outer Space Treaty represents the basic legal framework of international space law. Some of its principles are:

- The treaty also states that space shall be free for exploration and used by all the States.
- The treaty explicitly forbids any government to claim a celestial resource such as the Moon or a planet.
- The Treaty states that "outer space, including the Moon and other celestial bodies, is not subject to national appropriation by claim of sovereignty, by means of use or occupation, or by any other means". However, the State that launches a space object retains jurisdiction and control over that object.



MAJOR MISSIONS OF OTHER COUNTRIES

PARKER SOLAR PROBE

- Parker Solar Probe is a NASA spacecraft enroute to study the outer corona of the Sun.
- The probe was launched in August 2018.
- It will approach to within 6.2 million kilometers from the photosphere (surface) of the Sun and will travel, fast as 700,000 km/h in the orbit.
- The Parker Solar Probe will be the first spacecraft to fly into the outer solar corona.



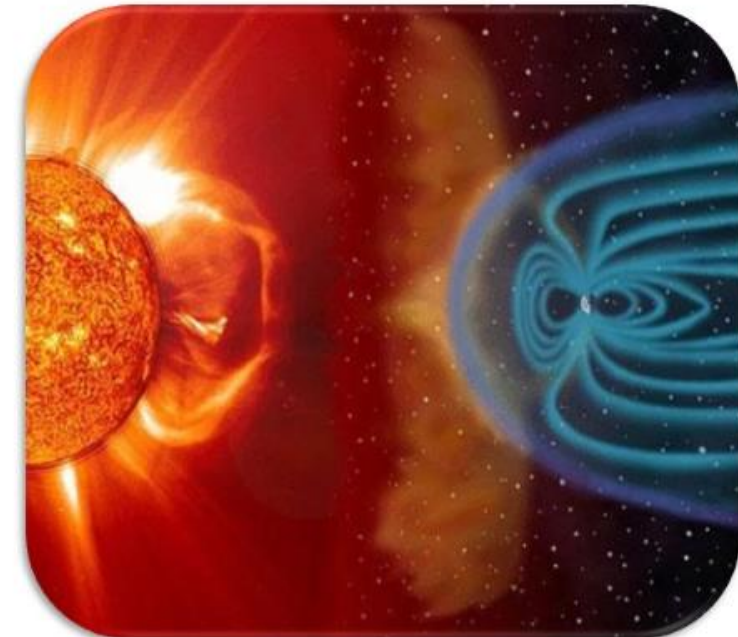
MAJOR MISSIONS OF OTHER COUNTRIES

PARKER SOLAR PROBE

- It will assess the structure and dynamics of the Sun's corona and magnetic field; it will study the energy flow that heats the solar corona and causes the *solar wind*.

SOLAR WIND

- The **solar wind** is a stream of charged particles released from the upper atmosphere of the Sun, Corona. This plasma consists of mostly electrons, protons and alpha particles. The particles can escape the Sun's gravity because of their high energy resulting from the high temperature of the corona.
- The solar wind is supersonic and reaches speeds of 250 to 750 kilometers per second.



MAJOR MISSIONS OF OTHER COUNTRIES

PARKER SOLAR PROBE

- Parker Solar probe will face immense heat and radiation. The spacecraft's systems are protected from the extreme heat and radiation by a solar shield.
- If the shield were not between the spacecraft and the Sun, the probe would be damaged and become inoperative within few seconds.



MAJOR MISSIONS OF OTHER COUNTRIES

WHY SUCH A PROBE?

- We are affected by the **Space Weather**.
- Understanding the physics of corona and solar wind will help us better manage the technological infrastructure of Earth.
- The probe will help us
 - ✓ Improve satellite communications
 - ✓ Power grid issues
 - ✓ Radiation exposure on airline flights and
 - ✓ Astronaut safety.



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